Single Slit Diffraction

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Aim of the Experiment

To study the intensity distribution of Fraunhofer diffraction pattern by a single slit and measure the width of the slit for a given wavelength of laser light.

Apparatus Required

(i) Optical Bench.
(ii) Laser Light Source (He-Ne 1mW).
(iii) Focusing lenses and adjustable slit.
(iv) Photometer (Optical base plate) & Screen
(v) Digital Multi-meter with connecting wires.

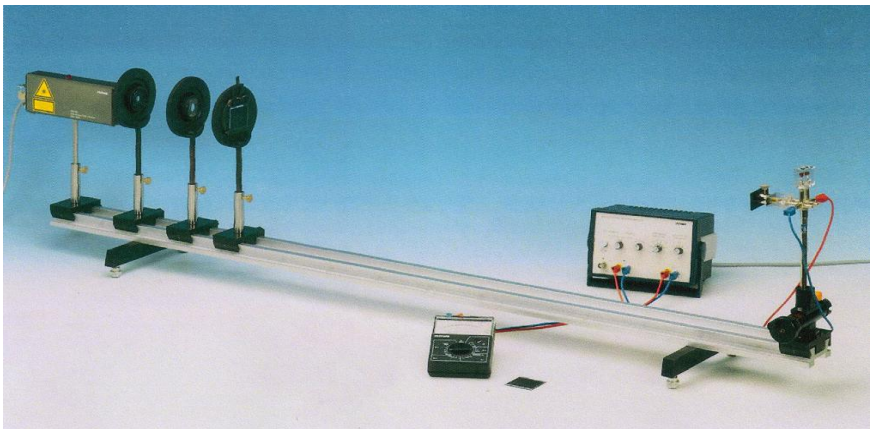


Fig-1: Experimental setup

Theory

When a beam of monochromatic light passes close to the edge of opaque obstacles or narrow slits it suffers deviation from its straight line path. This phenomenon becomes more interesting when the dimension of the slit or aperture is comparable to the wavelength of incident light. For such case, light not only bends towards the region of geometrical shadow but also shows non-uniform illumination outside the geometrical shadow with alternate variation of intensity. This is called as diffraction of light. There are two types of diffraction namely Fraunhofer type and Fresnel type. If the source of light and the screen are effectively at infinite distances from the slit, the diffraction is called as Fraunhofer diffraction which is usually achieved by the use of combination of convex lenses. Fig-1 depicts the general experimental set up for single slit diffraction.

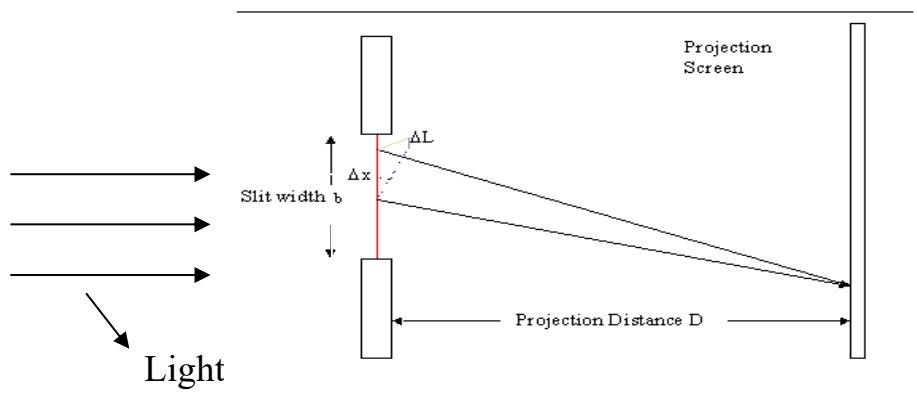


Fig-2: Schematic Diagram

Consider a beam of monochromatic light of wavelength 'λ' falls on the narrow vertical slit MN of width *'b'* normally as shown in Fig-2 so that a number of

alternate bright and dark fringes on both sides of central maximum are observed. The resultant intensity of the transmitted light can be expressed as

$$I = I_0 \frac{\sin^2 \beta}{\beta^2} \quad \text{where} \quad \beta = \frac{\pi b \sin \theta}{\lambda}$$

Where, θ = *angle of diffraction.*

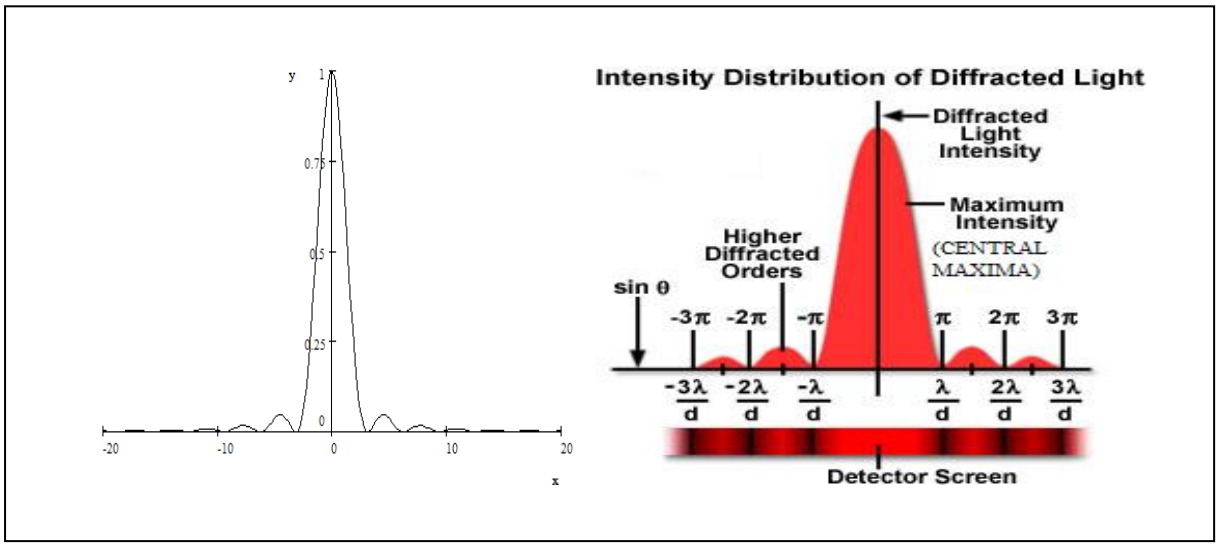


Fig-3: Typical intensity distribution of diffraction Pattern

The diffraction pattern consists of a principal maximum for $\beta = 0$, where all secondary wavelets arrive in phase, and several secondary maxima of diminishing intensity with equally spaced points of zero intensity at $\beta = m\pi$. The positions of minima of a single slit diffraction pattern are,

$$m\lambda = b \sin \theta, \quad m = \pm 1, \pm 2, \pm 3, \ldots\ldots$$

If θ is small i.e. the slit to screen distance D is large compared to the distance x_m between two minima (on either side of mth order maximum), then

$$\theta_{m-1} = \frac{x_{m-1}}{D}, \quad \theta_{m+1} = \frac{x_{m+1}}{D}$$

Now $b\,\theta_{m-1} = b\,\frac{x_{m-1}}{D} = (m-1)\,\lambda$ for (m-1)th minima.----------(1)

$b\,\theta_{m+1} = b\,\frac{x_{m+1}}{D} = (m+1)\,\lambda$ for (m+1)th minima.----------(2)

Now subtracting eqn(2) from eqn(1) we got,

$$b\frac{x_m}{D} = 2\lambda \text{ or } \lambda = b\,\frac{x_m}{2D}$$

From the above formula width of the slit '*b*' is calculated.

Procedure

(i) Mount the laser source and the screen on the opposite sides of the optical bench. Connect the laser to input A.C. signal.

(ii) Put the slit perpendicular to the beam of light mounted on a vertical stand in between the laser source and screen so that light coming from the laser passes through it and falls on the screen.

(iii) To obtain a parallel beam of light from the laser source, place the convex lenses (f= 20mm and f=100mm) in between the laser source and slit. Adjust the height and positions of the convex lenses from the laser and the slit so that bright and distinct fringes are obtained on the screen.

(iv) Remove the screen and place the photometer in this position so that the fringes will fall on its surface. Connect the photometer to digital multi-meter.

(v) Measure the vernier constants of the micrometer scale attached to the photometer.

(vi) Two or three secondary maxima should be obtained on either side of the central maximum on the photometer. Turn the circular scale attached at

the base of the cell in equal intervals and note the intensity of the fringes using multi-meter, scanning the whole diffraction pattern starting either from left end of the pattern to right end or vice versa.

(vii) Reach the other end of the fringe this way noting down the sufficient readings (approx. 70) of the intensities in between. Repeat the same procedure to take reading in reverse direction.

(viii) Measure the distance 'D' i.e. the separation between the slit and the photometer using the scale on optical bench.

(ix) Calculate angular deviation for all above positions and plot two separate graphs in θ vs. intensity, one for left to right and other for right to left observations. Calculate the slit width from the graph.

(x) Plot intensity distribution of the diffraction as a function of θ parallel to the plane of the slit. Find the width of given slit from the plot and compare this width with that measured by microscope.

Observations

Vernier constant of micrometer = 0.01mm

Wave length of the laser light = 633 nm

Distance of the slit from the photometer = 74.5 cm

Table-1: Intensity distribution of single slit pattern (x_0 corresponds to position of central maximum)

Sl. No.	Position (x cm)	$\theta_{lr} = (x-x_0)/D$	Photo current/voltage (mA/V) [Left to right]	Position (x cm)	$\theta_{rl} = (x-x_0)/D$	Photo current/voltage (mA/V) [Right to left]

1	0.02	-0.01812	2.4	1.12	-0.018792	1.1
2	0.07	-0.01744	2.3	1.17	-0.018121	1.2
3	0.12	-0.01679	2.1	1.22	-0.01745	1.1
4	0.17	-0.01611	1.6	1.27	-0.016779	1
5	0.22	-0.01544	1.0	1.32	-0.016107	0.9
6	0.27	-0.01476	0.5	1.37	-0.015436	0.6
7	0.32	-0.01409	0.4	1.42	-0.014765	0.4
8	0.37	-0.01342	0.7	1.47	-0.014094	0.6
9	0.42	-0.01275	1.6	1.52	-0.013423	1.0
10	0.47	-0.01208	3.0	1.57	-0.012753	2.1
					-0.012081	

Sl. No.	Position (x cm)	$\theta_{lr} = (x-x_0)/D$	Photo current/voltage (mA/V) [Left to right]	Position (x cm)	$\theta_{rl} = (x-x_0)/D$	Photo current/voltage (mA/V) [Right to left]
11	0.52	-0.01141	4.7	1.62	-0.011409	3.6
12	0.57	-0.01074	6.5	1.67	-0.011409	5.5
13	0.62	-0.01067	7.4	1.72	-0.010738	7.1
14	0.67	-0.009396	7.6	1.77	-0.009396	8.4
15	0.72	-0.008725	6.7	1.82	-0.008725	8.8
16	0.77	-0.008054	5.5	1.87	-0.008054	8.0

17	0.82	-0.007383	2.4	1.92	-0.007383	5.5
18	0.87	-0.006711	0.5	1.97	-0.006711	3.4
19	0.92	-0.006040	0.9	2.02	-0.00604	1.6
20	0.97	-0.005369	5.7	2.07	-0.005369	1.0
21	1.02	-0.004697	16.5	2.12	-0.004698	6.1
22	1.07	-0.004026	35.8	2.17	-0.004027	15.1
23	1.12	-0.003355	64.4	2.22	-0.003356	35.1
24	1.17	-0.002684	101.1	2.27	-0.002685	60.1
25	1.22	-0.002013	122.3	2.32	-0.002013	92.4
26	1.27	-0.001342	123.2	2.37	-0.001342	119.5
27	1.32	-0.0006711	123.8	2.42	-0.001342	121.4
28	1.37 $=x_0$	0	124	2.47	-0.000671	121.9
29	1.42	0.0006711	123.4	2.52	0	123.5

Sl. No.	Position (x cm)	$\theta_{lr} = (x-x_0)/D$	Photo current/voltage (mA/V) [Left to right]	Position (x cm)	$\theta_{rl} = (x-x_0)/D$	Photo current/voltage (mA/V) [Right to left]
30	1.47	0.001342	123.7	2.57	0.000671	122.3
31	1.52	0.002013	123.5	2.62	0.001342	122.2
32	1.57	0.002684	123	2.67	0.002013	122

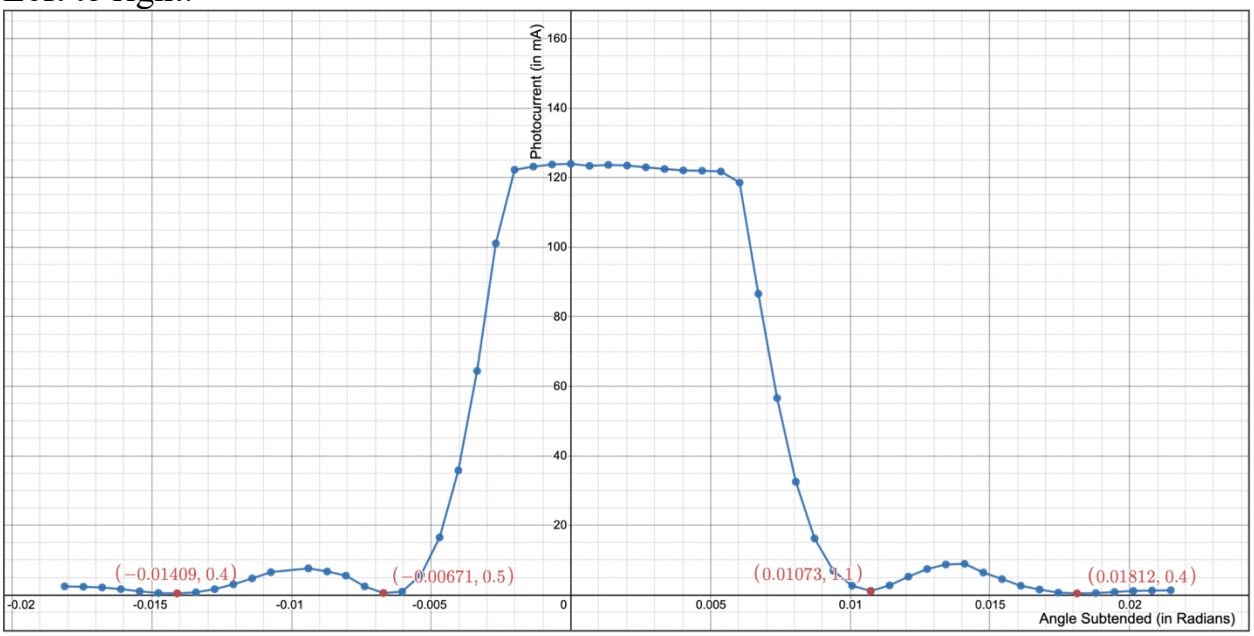
33	1.62	0.003355	122.5	2.72	0.002685	121.8
34	1.67	0.004026	122.1	2.77	0.003356	120.9
35	1.72	0.004647	122	2.82	0.004027	120.5
36	1.77	0.005369	121.8	2.87	0.004698	120
37	1.82	0.006040	118.6	2.92	0.005369	118.8
38	1.87	0.006711	86.6	2.97	0.00604	90.1
39	1.92	0.007382	56.6	3.02	0.006711	56.5
40	1.97	0.008053	32.5	3.07	0.007383	31.9
41	2.02	0.008724	16.2	3.12	0.008054	14.6
42	2.07	0.009345	6.9	3.17	0.008725	5.8
43	2.12	0.010067	2.0	3.22	0.009396	2.3
44	2.17	0.01073	1.1	3.27	0.010067	1.4
45	2.22	0.011404	2.7	3.32	0.010738	3.5
46	2.27	0.012081	5.2	3.37	0.011409	6.2
47	2.32	0.012751	7.4	3.42	0.012081	2.0
48	2.37	0.013422	8.7	3.47	0.012752	5.7

Sl. No.	Position (x cm)	$\theta_{lr} = (x-x_0)/D$	Photo current/voltage (mA/V) [Left to right]	Position (x cm)	$\theta_{rl} = (x-x_0)/D$	Photo current/voltage (mA/V) [Right to left]

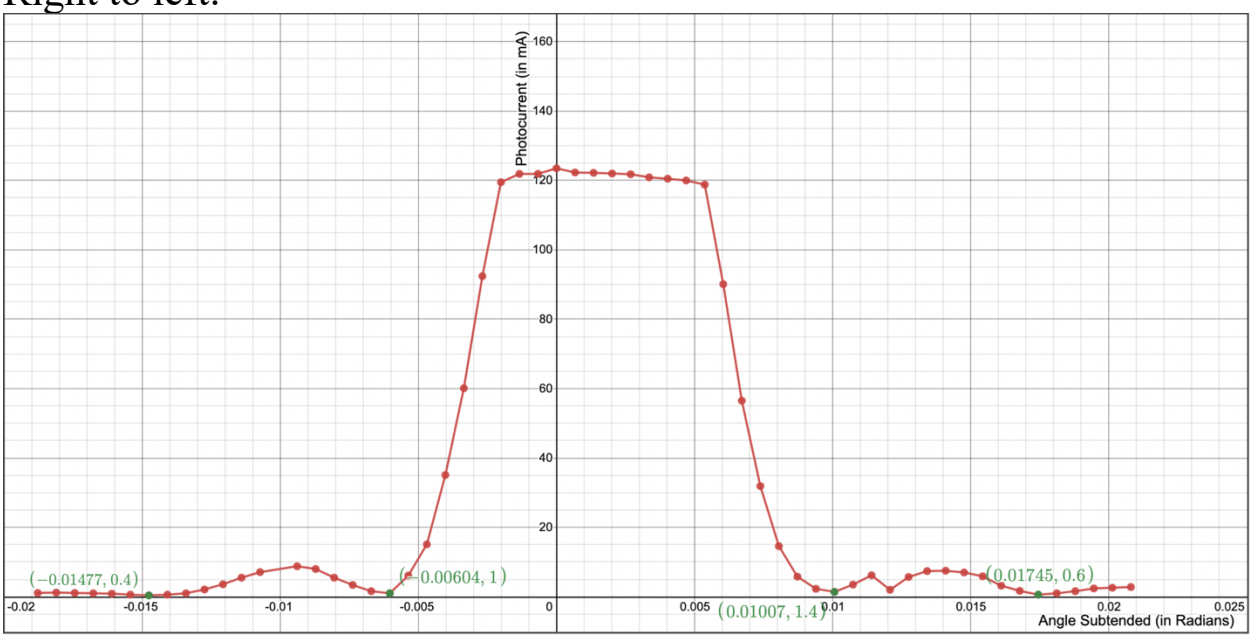
49	2.42	0.014094	8.9	3.52	0.013423	7.4
50	2.47	0.014765	6.4	3.57	0.014094	7.5
51	2.52	0.015436	4.5	3.62	0.014765	7.0
52	2.57	0.016107	2.6	3.67	0.015436	5.9
53	2.62	0.016778	1.5	3.72	0.016107	3.2
54	2.67	0.01744	0.6	3.77	0.016779	1.7
55	2.72	0.01812	0.4	3.82	0.01745	0.6
56	2.77	0.01879	0.5	3.87	0.018121	1.0
57	2.82	0.01946	0.8	3.92	0.019463	1.6
58	2.87	0.020134	1.1	3.97	0.019463	2.4
59	2.92	0.020805	1.2	4.02	0.020134	2.6
60	2.97	0.021476	1.3	4.07	0.020805	2.8

Graphs:

Left to right:



Right to left:



Calculations:

From the graphs we find $\Delta\theta$ where θ is the angle between two consecutive minima

From the first graph, (it must be kept in mind that the angle subtended by central fringe is twice $\Delta\theta$) we have,

$$\Delta\theta = \frac{(0.01812-(-0.01409)}{4} = \frac{0.03221}{4} = 0.008052 \text{ radians}$$

From the second graph, we have,

$$\Delta\theta = \frac{(0.01745-(-0.01477))}{4} = 0.008055 \text{ radians}$$

Hence $\Delta\theta = \frac{0.008052+0.008055}{2} = 0.008054$ radians

Since we know $\Delta\theta \times b = \lambda$, where $\lambda = 633$nm

We get $b = \frac{\lambda}{\Delta\theta} = \frac{633}{8.054}$ μm $= 78.5945$ μm

Calculation of error:

Since we know $b = \frac{\lambda}{\Delta\theta} = \frac{\lambda}{x/D} = \frac{\lambda D}{x}$

Assuming laser light to be highly monochromatic, so relative error in λ is 0.

Absolute error in x, D are $\Delta x = 0.01\text{mm} = 0.001\text{cm}$ and $\Delta D = 0.1\text{cm}$.

Here, $x = \Delta\theta \times D = 0.6\text{cm}$

Fractional error in b= $\frac{\Delta b}{b} = \frac{\Delta x}{x} + \frac{\Delta D}{D} = \frac{0.001}{0.6} + \frac{0.1}{74.5} = 0.003008 \approx 0$

Percentage error in $b = \frac{\Delta b}{b}.100 = 0.3\%$

Absolute error in $b = 0.2\mu\text{m}$.

Conclusions

(i) Diffraction pattern due to single slit aperture is observed. Wave nature of light is confirmed(Huygens's principle). From the experimental data obtained, the width of slit was found.. The graph of intensity vs angle subtended at screen was plotted and its similarity with theoretical plot was confirmed.

(ii) Some limitations faced by this experiment are need of monochromatic light of suitable wavelength: for instance, the experiment would not work for normal light because of formation of superposition of the pattern due to multiple wavelengths . Background lighting is also very important and it is important to do the experiment in a dark room so as to detect the fringes clearly.

(iii) The experiment explained here has various applications. To name a few: diffraction gratings(to find absorption and emission spectra of excited atomic species), as such diffraction effects are important in optical instruments like large telescopes, they must be kept in mind when designing telescopes, diffraction plays a vital role in astronomy, X-ray diffraction is used to study crystals at sub-atomic scales and has various other applications in modern science.

Precautions

(i) One should not look directly to LASER light because it may damage the eye.

(ii) The optical bench should be levelled properly.

Considerations

(i) What is diffraction of light?

When light form a source bends around the corners of obstructions in its path , the phenomenon of variation of its intensity is called diffraction.

(ii) How diffraction is different from interference?

Diffraction is not much different than interference fundamentally. More concretely speaking, they are governed by the same mathematical principle: Huygens's principle. It is mainly a matter of convention, when multiple coherent sources are present, it is usually called diffraction and for a few sources it is usually called interference. In certain analysis like young's double slit experiment we often tend to ignore the finite size of the slits to get rid of "self-interfering/diffraction effects" and just treat those as 2 point sources and call the phenomenon as interference. It is important to note that diffraction and interference always happen at the same time and there is no important physical difference between the two.

(iii) Name different types of diffraction and their properties.

There are 2 types of diffraction: Fresnel and Fraunhofer diffraction. In optics, the Fresnel diffraction equation for near-field diffraction is an approximation of the Kirchhoff-Fresnel diffraction that can be applied to the propagation of waves in the near field. It is used to calculate the diffraction pattern created by waves passing through an aperture or around an object, when viewed from relatively close to the object. Fraunhofer diffraction is the far field approximation(distances from different parts of the aperture to a point on the screen is approximately same)

(iv) What will happen to the intensity distribution if the width of the slit is increased or decreased?

When the width of slit is increased, the width of central and secondary fringes decreases and the intensity of the central maxima

increases due to more amount of light being let through the aperture (assuming that source illuminating the aperture is the same).

(v) What is the difference between single slit and double slit diffraction pattern?

The diffraction pattern due to the single slit diffraction pattern acts as an envelope for the 2 slit diffraction pattern. Mathematically the pattern looks like the intensity distribution of YDSE multiplied with that of single slit diffraction.